University of BRISTOL

Mathematical Exploration Project

Week 3 · Introduction

MATH30024 · Perspectives in Mathematics · Theme 3B

Aims of today

- Understand the requirements of the **Mathematical Exploration Project**.
- Discuss about **team-member types** within your group.
- Start to **choose a project topic** together.

The Mathematical Exploration Project

Learning outcomes:

- Independent learning of mathematics
- Learning from scholarly materials
- Effective group working
- Written communication skills
- Presentation skills

Define your **own** project in an area of mathematics or mathematics education that extends beyond the standard curriculum.

Define your own project

- You define the project — it is open-ended.
- Find sources in the **literature** (not YouTube/Wikipedia).
- Aim for the "wow factor" — make your project stand out.

Assessment

- **Group report**, 20–30 pages (lecturer + moderator) — 25%
- **Group presentation** (lecturer + peer assessment) — 15%
- **Individual reflection** group contributions and AI use — 10%

How the sessions run

- Sit in your **Theme 3B groups** on Thursdays.
- Each session: $\sim$10 minutes of material, then I speak with each group individually.
- Use the rest of the time to **work as a group** and raise questions about group dynamics early.

Schedule (weeks 3–11)

Wk	Focus	Deliverable
3	Project overview; topic selection	—
4	Assessment criteria; topic choice	Topic settled
5	Writing and referencing; group work	—
6	Consolidation week (no class)	—
7	Avoiding plagiarism; group work	—
8	—	Draft report
9	—	Feedback
10	Practice presentation	Final report
11	—	Assessed presentation

Choosing your project

- A topic in **mathematics** or **mathematics education**.
- Some ideas are on Blackboard.
- **Group decision:** how will you decide? (e.g. each member brings an option with a source, then the team votes.)

What area of mathematics excites you, and why? Can you connect it to a real-world problem? Is there a question, conjecture or paradox that sparks curiosity?

Making your project stand out

- How could you visualize your ideas? (e.g., dynamic graphs, interactive visualizations, 3D models, simulations)
- Can you use creative storytelling or narrative to present your project (video, animation, or story-based explanation)?
- Are there opportunities for interdisciplinary connections (math + physics, art, biology, social sciences, engineering)?
- Can you implement or code something original—even a small interactive tool or algorithm?

Working as a team (Belbin roles)

- Reflect on your Week 2 group: what worked well or badly, and why?
- What attributes did members bring that helped or hindered?
- Everyone has a **team role** with strengths and allowable weaknesses — know yours, and cover the gaps.

If your group hits problems

- Come and talk to me to explain the issue (e.g. a member not contributing).
- Raise issues early — not a week before the deadline.

Before next session

- Come to the next session ready to **choose and work on a topic**.
- Agree how your group will communicate and share work.

Let's get started.